La Scuola d'Italia / Huson / Physics Nickname:
23 October 2026

2.3 Homework: Constant Acceleration

Reference (provided). These hold when the acceleration a is constant.

$$v = v_0 + at \qquad v^2 = v_0^2 + 2a(x - x_0)$$

$$x = x_0 + v_0 t + \tfrac{1}{2}at^2 \qquad \bar{v} = \frac{v + v_0}{2}$$

For each problem, list what you know (v_0, v, a, t, $x - x_0$) and what you want, then choose the equation that connects them. Show the units.

1. The cart from 2.2 is released from rest on a tilted Vernier track. It speeds up at a constant $a = 0.500$ m/s^2.

   (a) An acceleration of 0.500 m/s^2 means the velocity grows by 0.500 m/s every second. Fill in the table.

t (s)	0	1.00	2.00	3.00	4.00
v (m/s)					

   (b) How far has the cart rolled at $t = 2.00$ s? In 2.2, Graphical Analysis measured 1.000 m. Does your answer agree?

2. Tesla advertises that its Model S Plaid goes from 0 to 60 mi/h in 1.99 s. Take 60 mi/h = 26.8 m/s and assume the acceleration is constant.

   (a) Find the car's acceleration.

   (b) How far does the car travel in those 1.99 s?

3. The UK Highway Code, the official rules for British drivers, says that a car at 30 mi/h (13.4 m/s) needs 14 m of braking distance to stop.

   (a) Find the car's acceleration while it brakes. Why is it negative?

   (b) At 60 mi/h (26.8 m/s), with the same acceleration, find the braking distance. Twice the speed gives how many times the braking distance?

4. The elevators in the Shanghai Tower, 632 m tall, reach a top speed of 20.5 m/s. Suppose an elevator starts from rest and speeds up at a constant 1.0 m/s^2, about as much as passengers find comfortable.

   (a) How long does it take to reach top speed?

   (b) How far does it rise while speeding up?

   (c) At the top it slows down at the same rate, so it needs the same distance to stop. What is the shortest ride on which the elevator reaches top speed?